FROM THE
EDITOR-IN-CHIEF

Group Captain Shubhanshu Shukla may be the second Indian astronaut to go into space, 41 years after Wing Commander Rakesh Sharma's feat in 1984. But Shukla's voyage on board the American spaceflight Axiom Mission 4 also marks a historic first step. He has become the first Indian to visit the International Space Station (ISS) and experience firsthand its 28,000 kmph orbit around the Earth. But there's more to it than just that. His ticket may have cost Rs 500 crore. Still, Shukla's planned two-week 'workation' aboard the ISS will yield primary data of inestimable value, for he is among the four Indian astronauts picked for the Gaganyaan Mission, India's ambitious human spaceflight programme. The experiential richness Shukla brings back from his hitchhike to the galaxy will go right into an all-Indian spaceflight manual. Future relevance is what gilds his feat: each step here is a launch pad for the next flight. What this heralds is a New Space Odyssey for India as it joins the heavyweights of human space exploration.

First, a tip of the hat to the real protagonist here: the Indian Space Research Organisation (ISRO), an excellent body that has made pioneering initiatives at a fraction of the cost borne by other nations. While ISRO spent the past decades focused on achieving self-sufficiency in launch vehicles and satellites for India's developmental needs, it is now expanding its orbit towards scientific exploration of our solar system. The next step on ISRO's programme itinerary is a three-member crewed flight in 2027, on a rocket made in India, as part of Gaganyaan. Its logistical and safety modalities will be ironed out with three uncrewed flights before we get to the human stage. The crewed flight, in its turn, is envisaged as an occasion to test, experiment and gain experience for the future.

▲ **April 30, 1984**

Humans in space need a big reboot to their physiological and psychological selves. From bare essentials like oxygen, pressure and temperature to complex functions, all facets of living call for adjustment. Everything about our cognitive, metabolic and motor apparatus is made for normal gravity. How we can keep ticking in its virtual absence is a huge part of orienting ourselves to outer space. As Shukla pointed out when he reached space, even basic human functions like having food have to be relearnt; the body has to master how to navigate the microgravity within a spacecraft before being able to manage the spacecraft's navigation systems. No amount of theory works without practice; there's no such thing as transferred wisdom. India will have to build its own first-hand knowledge base. The value that Shukla's inputs will have can be seen right here.

Gaganyaan's next step up the pyramid is an even more ambitious one. This is the project to build India's very own home in space, Bharatiya Antariksh Station (BAS), a low-orbit platform like the ISS. This is a complex task, set out in several stages. A space station is built out of distinct modules that go up separately over the years and get hitched to each other like wagons. The ISS took from 1998 to 2011 before all the components were fully assembled. ISRO's first BAS module is slated for a 2028 launch; its final piece will be in place by 2035. All the Indians who pitch tent on this little ledge in outer space will also be voyaging into pure science, continuing the experiments Shukla (a.k.a. 'Shux') is presently undertaking, testing the effects living in microgravity has on humans, vis-à-vis muscle atrophy, bone density loss, endurance et al.

After firing up young minds with the successful moon landing of Chandrayaan-3 in 2023, India has big plans for lunar exploration. Chandrayaan-4, scheduled for launch in 2027-28, will not just land on the moon—the spaceship will bring back lunar samples for our scientists to research on. India is also catching its breath for the big one. No human has been on the moon since 1972. Space ambitions had long diverted elsewhere; NASA plans to fill that blank by 2027. Russia is back in the race and China is making big strides too. All the space experience that lies ahead will prove invaluable, for India aims to join the club with a manned lunar mission by 2040.

Experience is exactly what Group Editorial Director Raj Chengappa brings to the subject of space. As a young journalist, he covered Rakesh Sharma's 1984 space voyage for INDIA TODAY. This week, he telescopes all of India's long space walk, its past and glittering future, and sets it in context. Our soaring space ambitions come at a time when there's a renewed global buzz about outer space. Human colonisation and habitation, Mars missions, all are on the radar. Private investment is peaking: Shux's ride was on a NASA-SpaceX joint venture. While ISRO is the rare PSU that actually works brilliantly, the private Indian ecosystem too is picking up, readying for a stage when private players can build, own and operate in space. Luckily, space also sees international cooperation on a scale invisible on earth. The ISS, a five-sided collaboration between the US, Russia, Europe, Canada and Japan, is a case in point. From the ISS vantage point, 400 km above the Earth, Shukla gets to see 16 sunrises and sunsets in a day. Here's wishing India many more sunrises.

(Aroon Purie)